

# HW04 Automation Testing Report

## Document control

| Field                      | Value                                                                                                                                 |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Student                    | Lê Tuấn Lộc                                                                                                                           |
| Student ID                 | 23127404                                                                                                                              |
| Class                      | CSC15003 (student-confirmed)                                                                                                          |
| Repository                 | <a href="https://github.com/KidCute1412/eshop-sut">https://github.com/KidCute1412/eshop-sut</a> (student-confirmed public repository) |
| Branch inspected           | 23127404-LeTuanLoc                                                                                                                    |
| Report status              | Nine-run execution evidence reconciled; manual submission artifacts pending                                                           |
| Final submission timestamp | <b>PENDING — ISO 8601</b>                                                                                                             |
| Execution revision         | 656991a598baf43cbed13b54bf1ac3f429c30f2 ; recorded in every selected report metadata file                                             |

## Executive summary

HW04 requires data-driven automation for the same three web features selected in HW02: FR-06 Product detail view, FR-10 Order state machine, and FR-12 Access control. Each feature requires at least 12 cases and execution on three browsers, producing at least nine attributable HTML reports overall.

The deliverables suite defines 51 external-data cases: 16 for FR-06, 16 for FR-10, and 19 for FR-12. All cases were scheduled on Chromium, Firefox, and WebKit in nine selected feature/browser runs. The nine complete HTML/JSON/metadata report directories passed structural identity, timestamp, case-count, completion, and totals validation. Across 153 browser attempts, 102 reached assertions: 64 passed and 38 failed across 19 unique logical cases. The other 51 attempts failed during Firefox `browserContext.newPage()` fixture setup before the test body could observe the SUT.

FR-06 produced 20 passes and 28 recorded failures across 48 executions; 16 failures were environmental and 12 were assertions. FR-10 produced 24 passes and 24 recorded failures across 48 executions; 16 were environmental and 8 were assertions. FR-12 produced 20 passes and 37 recorded failures across 57 executions; 19 were environmental and 18 were assertions.

Firefox was run headful after headless Firefox failed before `newPage()` on the assignment Windows environment. The headful final runs remained intermittently affected by `newPage()` timeouts; those failures are not counted as SUT defect candidates. The Markdown/PDF report set is complete. No public HW04 Issue URL or demo video is currently verified, and the four-day Git-history span remains noncompliant.

The full HW04 Git history contains nine qualifying test-script commits across 27 July and 9 August 2026, so the commit-count minimum is met. The separate four-calendar-day requirement remains noncompliant. This report does not infer, manufacture, or backdate missing history.

# 1. Scope and test oracle

| Pool | Requirement                 | Observable obligation                                                                                                                                                                                         | Source continuity                                 |
|------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| A    | FR-06 — Product detail view | Show image, name, formatted price, description, and category; accept only positive integer quantity with minimum 1; provide visible add-to-cart feedback                                                      | HW02 FR-06 design and root <code>README.md</code> |
| B    | FR-10 — Order state machine | Enforce permitted transitions across <code>pending</code> , <code>confirmed</code> , <code>shipping</code> , <code>delivered</code> , and <code>canceled</code> ; preserve final states and role restrictions | HW02 FR-10 design and root <code>README.md</code> |
| C    | FR-12 — Access control      | Restrict Admin UI, <code>/api/admin/*</code> , and data-changing product/category/coupon APIs to a valid JWT with <code>role = 'admin'</code>                                                                 | HW02 FR-12 design and root <code>README.md</code> |

The final 51-row matrix is in `../supporting-materials/test_case_matrix.md`, with IDs `FR06-01 – FR06-16`, `FR10-01 – FR10-16`, and `FR12-01 – FR12-19`. It maps every actual external-data row to its spec and to the Chromium, Firefox, and WebKit outcome/report path.

## 2. Test design and traceability

### 2.1 Coverage targets

| Feature | Planned IDs                           | Positive                                        | Negative/boundary/edge focus                               | Automated now         | Final evidence                                                             |
|---------|---------------------------------------|-------------------------------------------------|------------------------------------------------------------|-----------------------|----------------------------------------------------------------------------|
| FR-06   | Source <code>FR06-01 – FR06-16</code> | Existing products and valid quantity boundaries | Content/format, quantity constraints, feedback, unknown ID | 16 cases × 3 browsers | 48 attempted; 32 reached assertions: 20 pass, 12 fail; 16 fixture failures |
| FR-10   | Source <code>FR10-01 – FR10-16</code> | Permitted admin/user transitions                | Skipped transitions, role limits, and final-state exits    | 16 cases × 3 browsers | 48 attempted; 32 reached assertions: 24 pass, 8 fail; 16 fixture failures  |
| FR-12   | Source <code>FR12-01 – FR12-19</code> | Authorized Admin UI/API operations              | Guest/user/malformed authorization and protected mutations | 19 cases × 3 browsers | 57 attempted; 38 reached assertions: 20 pass, 18 fail; 19 fixture failures |

### 2.2 Required row-level traceability

Every final case must map through the following chain:

Requirement → case ID → external data ID → test title/spec path → browser run ID → HTML report/evidence → defect/Issue when failed

The final matrix must record concrete preconditions, test data, steps, expected result, priority, automation decision, script path, and evidence reference. A case marked `Fail` must reference a defect or a documented non-SUT failure classification.

## 3. Automation implementation

### 3.1 Current inventory

| Feature | Spec file                                            | External data                                         | Data-driven proof                | Assertion patterns                                             | Setup/cleanup                                   | State                           |
|---------|------------------------------------------------------|-------------------------------------------------------|----------------------------------|----------------------------------------------------------------|-------------------------------------------------|---------------------------------|
| FR-06   | ../automation/tests/fr06-product-detail.spec.ts      | ../automation/test-data/fr06-product-detail.json      | loadCases(..., 12) loads 16 rows | Text/content, visibility, count, attribute, value/validity     | View cases plus cart-state-sensitive actions    | Attempted in 3 selected reports |
| FR-10   | ../automation/tests/fr10-order-state-machine.spec.ts | ../automation/test-data/fr10-order-state-machine.json | loadCases(..., 12) loads 16 rows | HTTP status/body, persisted state, visibility and button count | API driver arranges a fresh order state per row | Attempted in 3 selected reports |
| FR-12   | ../automation/tests/fr12-access-control.spec.ts      | ../automation/test-data/fr12-access-control.json      | loadCases(..., 12) loads 19 rows | HTTP status/content type/body plus Admin/Login UI visibility   | Disposable mutations with cleanup helper        | Attempted in 3 selected reports |

### 3.2 Current Playwright configuration

The deliverables suite pins Playwright 1.55.0 and defines Chromium, Firefox, and WebKit projects. It uses Playwright web-first assertions, workers: 1, CI-only retry, screenshots/traces/videos on failure, and a custom metadata reporter. Its report key includes feature, browser, and an ISO-derived timestamp; metadata includes Run by: 23127404, feature, browser, and timestamp.

The legacy bundle outside deliverables still produces one combined playwright-report directory. The new run:matrix workflow is intended to preserve distinct feature/browser report directories under the automation workspace. Each generated directory must still be opened and verified before it enters the final manifest.

### 3.3 Final implementation inventory template

| Feature | Case count     | Spec file(s)                                    | Data file(s)                                     | Locator strategy                                              | Assertions                                         | Isolation/reset                 | Reviewer approval                                              |
|---------|----------------|-------------------------------------------------|--------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------|---------------------------------|----------------------------------------------------------------|
| FR-06   | 16 source rows | ../automation/tests/fr06-product-detail.spec.ts | ../automation/test-data/fr06-product-detail.json | Roles plus scoped CSS where no stable semantic locator exists | Text, visibility, count, attribute, value/validity | Executed serially on 3 browsers | Human evidence triage recorded; final student sign-off pending |

| Feature | Case count     | Spec file(s)                                         | Data file(s)                                          | Locator strategy                                   | Assertions                                     | Isolation/reset                   | Reviewer approval                                       |
|---------|----------------|------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------|------------------------------------------------|-----------------------------------|---------------------------------------------------------|
| FR-10   | 16 source rows | ../automation/tests/fr10-order-state-machine.spec.ts | ../automation/test-data/fr10-order-state-machine.json | API arrangement plus exact order-cell/row locators | Status/body, persisted state, visibility/count | Fresh arranged order per row      | Substring locator corrected and all FR-10 runs repeated |
| FR-12   | 19 source rows | ../automation/tests/fr12-access-control.spec.ts      | ../automation/test-data/fr12-access-control.json      | API assertions plus Admin/Login headings           | Status, content type/body, visibility          | Cleanup helper for created probes | Executed; access-control failures retained              |

## 4. Execution and multi-browser evidence

### 4.1 Environment record

| Field                        | Final value                                                                                                                    |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Operating system             | Windows 10 Home Single Language; Windows NT 10.0.26200.0                                                                       |
| Node.js / npm                | Node.js 22.17.0 ; npm 10.9.2                                                                                                   |
| Playwright                   | 1.55.0                                                                                                                         |
| Browser modes/versions       | Chromium 140.0.7339.16 headless; Firefox 141.0 headful; WebKit 26.0 headless                                                   |
| Frontend URL                 | Current default http://localhost:5173 ; <b>confirm final URL</b>                                                               |
| Backend URL                  | Current default http://localhost:3000 ; <b>confirm final URL</b>                                                               |
| Execution revision/data seed | 656991a598baf43cbcd13b54bf1ac3f429c30f2 ; reports generated from clean seed database and metadata records the revision/command |
| Execution operator           | Lê Tuấn Lộc — 23127404                                                                                                         |

### 4.2 Historical repository evidence — excluded from final totals

| Evidence             | Repository-recorded value                                               | Qualification                                                 |
|----------------------|-------------------------------------------------------------------------|---------------------------------------------------------------|
| Logical case         | FR06-DT-01                                                              | One view-only FR-06 case                                      |
| Browsers represented | Chromium, Firefox, WebKit                                               | Stored in one combined report                                 |
| Recorded timestamp   | 2026-07-27T08:11:08.301Z                                                | Taken from the existing student report/report metadata        |
| Recorded result      | 0 passed, 3 failed                                                      | Report says each browser failed at the category assertion     |
| Preserved report     | ../schools/23127404-automation/playwright-report/index.html plus assets | Historical only; not copied into the final nine-run structure |
| Defect status        | Candidate only for this HW04 package                                    | No verified HW04 public Issue URL recorded here               |

The preceding row is provenance-aware reporting of an existing artifact. It does not establish that the environment remains unchanged or that the current SUT still exhibits the behavior.

### 4.3 Final execution ledger

| Run ID       | Feature/browser                | ISO timestamp            | Attempted           | Passed    | Recorded failed | Environment failures within failed | Selected report directory                                  |
|--------------|--------------------------------|--------------------------|---------------------|-----------|-----------------|------------------------------------|------------------------------------------------------------|
| RUN-A-CHR    | FR-06 / Chromium               | 2026-08-09T17:11:54.712Z | 16                  | 10        | 6               | 0                                  | automation/reports/fr06-chromium-2026-08-09T17-11-54-712Z/ |
| RUN-A-FF     | FR-06 / Firefox                | 2026-08-09T17:13:19.066Z | 16                  | 0         | 16              | 16                                 | automation/reports/fr06-firefox-2026-08-09T17-13-19-066Z/  |
| RUN-A-WK     | FR-06 / WebKit                 | 2026-08-09T17:16:10.035Z | 16                  | 10        | 6               | 0                                  | automation/reports/fr06-webkit-2026-08-09T17-16-10-035Z/   |
| RUN-B-CHR    | FR-10 / Chromium               | 2026-08-09T17:17:32.516Z | 16                  | 12        | 4               | 0                                  | automation/reports/fr10-chromium-2026-08-09T17-17-32-516Z/ |
| RUN-B-FF     | FR-10 / Firefox                | 2026-08-09T17:18:43.660Z | 16                  | 0         | 16              | 16                                 | automation/reports/fr10-firefox-2026-08-09T17-18-43-660Z/  |
| RUN-B-WK     | FR-10 / WebKit                 | 2026-08-09T17:21:22.978Z | 16                  | 12        | 4               | 0                                  | automation/reports/fr10-webkit-2026-08-09T17-21-22-978Z/   |
| RUN-C-CHR    | FR-12 / Chromium               | 2026-08-09T17:22:38.418Z | 19                  | 10        | 9               | 0                                  | automation/reports/fr12-chromium-2026-08-09T17-22-38-418Z/ |
| RUN-C-FF     | FR-12 / Firefox                | 2026-08-09T17:28:39.697Z | 19                  | 0         | 19              | 19                                 | automation/reports/fr12-firefox-2026-08-09T17-28-39-697Z/  |
| RUN-C-WK     | FR-12 / WebKit                 | 2026-08-09T17:27:20.694Z | 19                  | 10        | 9               | 0                                  | automation/reports/fr12-webkit-2026-08-09T17-27-20-694Z/   |
| <b>Total</b> | <b>3 features / 3 browsers</b> | —                        | <b>153 attempts</b> | <b>64</b> | <b>89</b>       | <b>51</b>                          | <b>Nine structurally validated directories</b>             |

The 89 runner failures equal 38 assertion failures plus 51 Firefox environment failures. All nine selected HTML/JSON/metadata report sets passed structural identity, timestamp, case-count, completion, report/data-directory presence, and totals validation. Each selected report metadata file records the exact `node scripts/run-matrix.mjs` command and revision `656991a598baf43cbcd13b54bf1ac3f429c30f2`.

The selected reports were generated from revision `656991a598baf43cbcd13b54bf1ac3f429c30f2`; the revision is embedded in every report metadata file and HTML attribution banner.

## 5. Human review and AI gap analysis

Human review retained strict requirement assertions, corrected one demonstrated locator defect, reran FR-10, and separated Firefox infrastructure symptoms from SUT behavior.

| Feature/case                     | AI proposal or risk                                           | Why insufficient                                                                                   | Human correction visible/documented                                                                                              | Evidence state                                                                     |
|----------------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| FR-06 / FR06-07                  | A happy-path check could omit category                        | FR-06 explicitly requires category; a reduced oracle could create a false pass                     | Exact category assertion was retained                                                                                            | Failed by assertion in all three final browser runs                                |
| FR-06 / locators                 | CSS selectors could couple tests to styling                   | Classes are not stable user-facing contracts                                                       | Roles, accessible names, and exact text are preferred; scoped CSS remains only for price styling where semantic hooks are absent | Visible in final spec and reports                                                  |
| FR-06 / data                     | Inline literals would violate data-driven requirement         | Inline test arrays/objects are explicitly disallowed                                               | Input and expected values were moved to external JSON                                                                            | Visible in current spec/data                                                       |
| FR-06 / waits                    | Fixed sleeps could hide timing defects and cause flakiness    | Duration is environment-dependent                                                                  | Playwright assertions and auto-waiting are used                                                                                  | Visible in current spec                                                            |
| FR-10 / FR10-02                  | Initial row filtering used substring matching for an order ID | Order #2 could match rows such as #20, producing a false WebKit failure                            | Filter now requires an exact table cell for #<orderId>; all three FR-10 runs were repeated                                       | FR10-02 no longer fails in the selected final Chromium, Firefox, or WebKit reports |
| Firefox environment              | A browser failure could be misreported as a product defect    | Headless Firefox failed before <code>newPage()</code> ; headful runs still had page-setup timeouts | Firefox configured headful; 51 <code>newPage()</code> timeouts classified as environment failures                                | Final Firefox JSON reports: 16 FR-06, 16 FR-10, 19 FR-12 environment failures      |
| FR-12 / authorization assertions | Passing over status mismatches would hide security behavior   | FR-12 requires valid admin JWT and role for protected endpoints                                    | Expected 401/403 assertions were retained; actual 200/400 responses remain failures                                              | 18 assertion failures across nine logical cases in the selected reports            |

The mandatory critique is in `ai_critique.md`. The interaction record is in `ai_audit_report.md`.

## 6. Defects and issue tracking

The final runs produced 38 assertion failures across 19 unique logical cases: six FR-06, four FR-10, and nine FR-12 cases. These are documented as runtime-observed candidates in `./bugs/bug_report.md`; the 51 Firefox `newPage()` failures are documented separately as environment failures. No candidate is represented as a filed GitHub defect because no verified public HW04 Issue URL or attached public screenshot has been supplied.

For every final failure:

1. complete case-level triage and rule out any remaining locator, assertion, data, authentication, state-reset, browser-installation, or infrastructure fault;
2. retest once when safe;
3. preserve the report, trace, and authentic screenshot;
4. create one deduplicated GitHub Issue for the root behavior; and
5. reconcile case IDs, run IDs, evidence references, and Issue URL in both bug files.

Prior HW02/HW03 issues may support regression context but are not evidence that a new HW04 Issue was filed or verified.

## 7. AI use

Required declaration: **I use AI tools for the following tasks**, preparing automation and submission documentation, reviewing traceability and evidence integrity, and drafting structured templates. The surviving audit history is incomplete: one historical exact prompt was not retained, and that omission is disclosed rather than reconstructed. See `ai_audit_report.md`.

## 8. Git history

The exported log in `../git/23127404_HW04_git_commit_log.txt` contains nine qualifying `.spec.ts` commits: legacy `f905546` on 27 July 2026 plus eight deliverables-suite commits from `f732b2c` through `aa316e0` on 9 August 2026. The commit minimum is met across two calendar days. The separate four-day requirement remains **noncompliant**; documentation commits do not change the qualifying span, and the shortfall must not be hidden through backdating or altered timestamps.

## 9. Demonstration videos and Agent Skill

| Item        | Requirement                                                                                                                                 | Current status                                          | Evidence                                                                                                        |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Main demo   | Unlisted YouTube, at least 5 minutes, Vietnamese narration, end-to-end script, multi-browser run/report, one human fix, authorship evidence | Not recorded                                            | <code>../video/demo_video.md</code>                                                                             |
| Agent Skill | Reusable data-driven multi-browser workflow plus separate end-to-end video                                                                  | Source present; end-to-end validation/demo not verified | <code>../agent-skill/playwright-data-driven-multibrowser/</code> ;<br><code>../video/agent_skill_demo.md</code> |

The Agent Skills assessment category is currently claimed as 0/10. Complete source exists, but this must change only after a valid end-to-end demonstration is recorded and verified.

## 10. Limitations and remaining work

- The 51 cases have qualifying selected reports, but the exact execution commit and verbatim per-run shell commands were not embedded/retained. Environment and browser versions are recorded.
- The 19 assertion-failing cases require final root-cause deduplication, public GitHub Issues, and authentic Issue screenshots before they can be claimed as submitted defects.
- Firefox produced 51 `newPage()` environment failures even in headful mode; those cases were not observed by the SUT assertions.
- Historical and current AI logs still require complete prompt/output coverage for every generation/review interaction.
- Environment and browser versions are recorded; the exact execution revision remains unavailable because it was not embedded in the selected reports.
- No HW04 public Issue URL or final defect screenshot is verified.

- Both video links are absent; the main demo is mandatory, and Agent Skill points cannot be claimed from source alone.
- Historical AI logging lacks at least one exact prompt and timestamp.
- Nine qualifying commits are present across two days; the four-calendar-day history requirement remains unsatisfied.
- Markdown-to-PDF exports are complete and text-validated; the final archive must be regenerated after student-controlled URLs and identity fields are supplied.

## 11. Final reconciliation table

This table is reconciled from the nine selected final JSON reports and their environment triage.

| Measure                                       | FR-06 | FR-10 | FR-12 | Total |
|-----------------------------------------------|-------|-------|-------|-------|
| Designed logical cases                        | 16    | 16    | 19    | 51    |
| Automated logical cases                       | 16    | 16    | 19    | 51    |
| Browser attempts                              | 48    | 48    | 57    | 153   |
| Attempts reaching assertions                  | 32    | 32    | 38    | 102   |
| Passed assertions                             | 20    | 24    | 20    | 64    |
| Recorded runner failures                      | 28    | 24    | 37    | 89    |
| Environment failures within recorded failures | 16    | 16    | 19    | 51    |
| Assertion failures                            | 12    | 8     | 18    | 38    |
| Unique assertion-failing logical cases        | 6     | 4     | 9     | 19    |
| Verified public HW04 Issues                   | 0     | 0     | 0     | 0     |

The arithmetic reconciles:  $64 + 38 + 51 = 153$ ,  $64 + 38 = 102$ , and  $38 + 51 = 89$ . Issue counts must be updated only from verified public URLs.